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Healthy broilers disseminate antibiotic resistance in response to tetracycline input in feed concentrates

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ABSTRACT

Wide varieties of antibiotics are used in poultry farms to improve the growth and also to control the infection in broiler chicken. To identify the seriousness of the same in the poultry sector, current study has been designed to analyze the presence of tetracycline in poultry feed and also the tetracycline resistance among the bacteria released through the excreta of poultry. In the study, 27 bacteria belonging to the *E. coli* and *Klebsiella* sp. were isolated from the faecal samples collected from five different farms. Antibiotic susceptibility analysis showed 77% of *E. coli* and 100% of the *Klebsiella* sp. to be resistant to tetracycline. Further, molecular screening for *tetA* and *tetB* genes showed 85.18% of isolates to have *tetA* and 22.22% with *tetB*. The presence of tetracycline in collected feed samples was also analyzed quantitatively by Liquid chromatography-mass spectrometry (LC-MS). Here, three out of five feed samples were found to be positive for tetracycline. The study showed a direct correlation between the antibiotic supplemented feed and the emergence of antimicrobial resistance among the intestinal microflora. The results of the study indicate the need for strict control over antibiotic use in animal feed to limit the rapid evolution and spread of antimicrobial resistance.

Key words: Tetracycline resistance, broiler chicken, *Klebsiella pneumoniae*, *Escherichia coli*, Feed additives.

INTRODUCTION

Antimicrobial resistance (AMR) is one of the most concerning global issues affecting the health and economy adversely. The rapid emergence of resistance has been considered to be due to the unscientific use of antimicrobial agents in the health care, livestock and agriculture. The uncontrolled use of antibiotics in livestock farms is an alarming issue worldwide. As the microorganisms can develop resistance either by mutations or sharing the genes from resistant strains, the resistance spread poses serious health risk to the society [1]. Wide range of antibiotics have been supplemented through the feed of the cattle, poultry and pigs for the growth enhancement as well as prophylaxis against mycoplasma, protozoa and bacterial infections[2].

Tetracycline class of antibiotics are especially used in the animal feed for the prevention of diseases caused by *Mycoplasma*, *Pasteurella multocida* and *Escherichia coli*. However, bacteria are known to have variety of mechanisms like efflux pumps, enzymatic inactivation, ribosomal protection and target modification to resist the tetracycline. The situation is very significant with respect to the broiler chicken as it is a major source of meat around the world. Broilers are the fast growing chicken breeds which can give yield up to 2 kg within in a short period of 37 to 40 days and hence antibiotic use is common in this sector[3]. The growth enhancement properties of antibiotics such as chlortetracycline and oxytetracycline also favour their increased commercial use in animal feeds especially for the broiler chicken [2]. The antibiotic input is considered to cause drastic changes in the normal poultry gut microbiome with positive selection of drug resistant bacteria. This could result in the domination of resistant strains and hence have the high risk of environmental spread[4]. As the poultry faeces are used as potential manure for agricultural practices, the resistant strains can easily spread to the environment and contaminate the food and water chain[5]. Here, the environmentally released resistant bacteria can ultimately make the currently used antibiotics for humans to be ineffective[6].

The development of tetracycline resistance in bacteria is mainly by two different methods such as efflux pumps and ribosomal protection proteins. *TetA*, *tetB*, *tetD* *tetE* and *tetG* are the commonly found resistance genes in gram-negative bacteria. Efflux pumps contributes resistance to tetracycline through constantly reducing the antibiotic concentration inside the cell. *tetA* and *tetB* are the commonly found tetracycline resistance genes in the livestock associated

enterobacteriaceae[7]. The resistant bacteria having these genes can easily share it to a susceptible one through horizontal gene transfer. Conjugation is one of the efficient gene transfer methods in which plasmids are transferred to other strains[8,9]. There are broad spectrum plasmids which can transfer the resistant genes among the bacteria from different species[10].

In this study, the prevalence of antimicrobial resistance in two Gram-negative bacteria from the healthy poultry faeces has been analysed to identify its linkage to antibiotic supplemented feed.

MATERIALS AND METHODS

Sample collection

Commercial broiler farms from Ernakulum, Kerala, India were selected for the study as the district is very active poultry production. The broilers were females with an age range of 15 to 41 days and the approximate bird population was above 5000/farm. Feed and water for the birds were provided in an *ad-libitum* basis and the birds were healthy without any history of illness. For the study, 25 faecal samples and 5 feed samples were collected from 5 different farms. All the samples were collected aseptically and transferred immediately to the laboratory.

Extraction and quantification of tetracycline from feed by LC-MS/MS

From the collected feed samples, 2 grams were homogenized and into this 20 mL of the extraction solvent (98 mL Methanol and 2 mL HCl) was added. This was further mixed for 20 min at 200rpm in a shaker and centrifuged at 2500 rpm. The supernatant collected was filtered through 0.22 μ m sterile syringe filter and used for further analysis. Standard based quantification of tetracycline from these samples was done using Shimadzu LC-MS/MS 8045 System[11].

Isolation and identification of *Escherichia coli* and *Klebsiella* sp.

From the collected faeces samples, bacterial isolation was made on Mac Conkey agar and selected the lactose fermenting and mucoid colonies after overnight incubation. *E. coli* were further confirmed by culturing on Eosin Methylene Blue (EMB) agar followed by observation for the green metallic sheen. All the isolates were then cultured in Luria-Bertani broth for DNA isolation using Macherey Nagel NucleoSpin® Microbial DNA isolation kit (Cat. No.

740235.50). The isolated genomic DNA was checked by agarose gel electrophoresis and was further used for the 16S rDNA amplification using the universal primers 16SF (5'-AGAGTTTGATCMTGGCTC-3') and 16SR (5'-AAGGAGGTGWTCCARCC-3') (Jasim et al. 2014). The 50µL PCR reaction contained 50ng of DNA, 10 pm of forward and reverse primers each, 25 µL of 2X PCR master mixes and 19 µL sterile MiliQ water. The PCR was done on Agilent SureCycler 8800 and the reaction started with an initial denaturation at 94 °C for 3 min followed by 35 cycles of denaturation at 94 °C for 30 s, annealing at 58 °C for 30 s, and extension at 72 °C for 2 min with a final extension at 72 °C for 7 min. The amplicons obtained were confirmed to have 1500bp size range by agarose gel. The PCR products were further used for Sanger sequencing and the sequences were analysed using Basic Local Alignment Search Tool (BLAST) of NCBI database. The sequences were further used for the phylogenetic analysis using Maximum likelihood method with 1000 bootstraps by MEGA-X[12]. The 16S reference sequences of *K. pneumoniae* and *E. coli* were collected from NCBI database and used for phylogenetic analysis.

Antibiotic Susceptibility Test

As per the Clinical Laboratory Standards Institute (CLSI) guidelines, all the selected isolates were subjected to antibiotic susceptibility test (AST). For this, the isolates were inoculated into Muller Hinton broth and incubated for 12 hrs. The turbidity of the broth was further made equivalent to 0.5 McFarland standard and was used for inoculation over Muller Hinton agar plate. Then various antibiotic discs (Chloramphenicol-30mcg, Ciprofloxacin-5mcg, Gentamycin-10mcg, Tetracycline-30mcg, Ertapenem-10mcg, Imipenam-10mcg, Aztreonam-50mcg, Piperacillin/tazobactam-100/10mcg and Ceftriaxone-10mcg) were placed over the inoculated surface and the plates were incubated for 24 hr at 37°C to observe the zone of inhibition [13,14].

PCR Amplification of Tetracycline Resistance gene

The isolated genomic DNA was also used to amplify the *tetA* and *tetB* genes coding for resistance to tetracycline. The previously published primers were used for the amplification of *tetA* were (*tetA*-F) 5'-GGTTCACCTCGAACGACGTCA-3' and (*tetA*-R) 5'-CTGTCCGACAAGTTGCATGA-3'. The same for *tetB* was (*tetB*-F) 5'-CCTCAGCTTCTCAACGCGTG-3' and (*tetB*-R) 5'-GCACCTTGCTGATGACTCTT-3' [15].

The PCR was programmed for 30 cycles after initial denaturation at 94°C for 30 seconds followed by; 94 °C at 30 seconds, 62°C for 30 seconds and 72°C for 30 seconds for *tetA* and 94 °C for 30 seconds, 58 °C for 1 min and 72 °C for 1 min for *tetB* with a final extension at 72°C for 5 minutes. Amplified samples were analyzed on 1.2% agarose gel. The products obtained were further outsourced for Sanger sequencing at AgriGenome Labs Pvt Ltd., Kochi and the sequences were analysed using BLASTn in The comprehensive antimicrobial resistance database (CARD-<https://card.mcmaster.ca/>)[16].

RESULTS

Extraction and quantification of tetracycline from feed using LC-MS/MS

Among the 5 feed samples, 3 were detected positive for tetracycline (m/z 445.10>410.15) by LC-MS analysis. The quantification method was optimised by using 5 different concentrations (10, 20, 40, 80 and 100) of tetracycline reference standard. The linearity of calibration curve was excellent with a high correlation coefficient ($r^2=0.9993$). By comparative quantitative analysis, the concentrations of tetracycline present were calculated to be 8.55ppb, 5.848ppb and 0.183ppb for Farm 1, Farm 4 and Farm 5 respectively (Fig. 1). The final concentration was obtained by calculating the dilution factor.

$$\text{Residue/gram} = (\text{concentration/weight of the sample}) \times \text{volume of the solvent used}$$

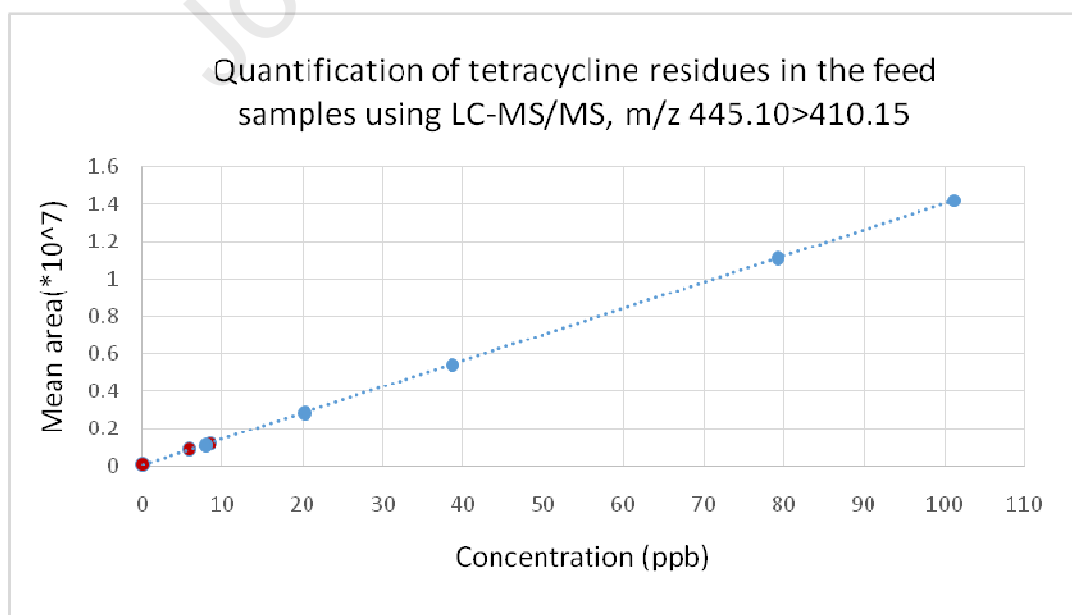


Fig. 1. Quantification of tetracycline residue in collected poultry feed samples using LC-MS/MS. Blue dots represent the concentration of standards and red dot represents the test samples.

Isolation and identification of *Escherichia coli* and *Klebsiella* sp.

A total of 22 isolates of *Escherichia coli* and 5 isolates of *Klebsiella pneumoniae* were obtained from the faeces samples and were confirmed by 16S rDNA sequence analysis. The sequences of *E. coli* and *K. pneumoniae* were submitted in Genbank with accession numbers MT476936-MT476958 and MT476959-MT476963 respectively. Here, NCBI BLASTn analysis confirmed the isolates as *E. coli* and *Klebsiella* sp. and was supported by the results of phylogenetic analysis (Fig. 2).

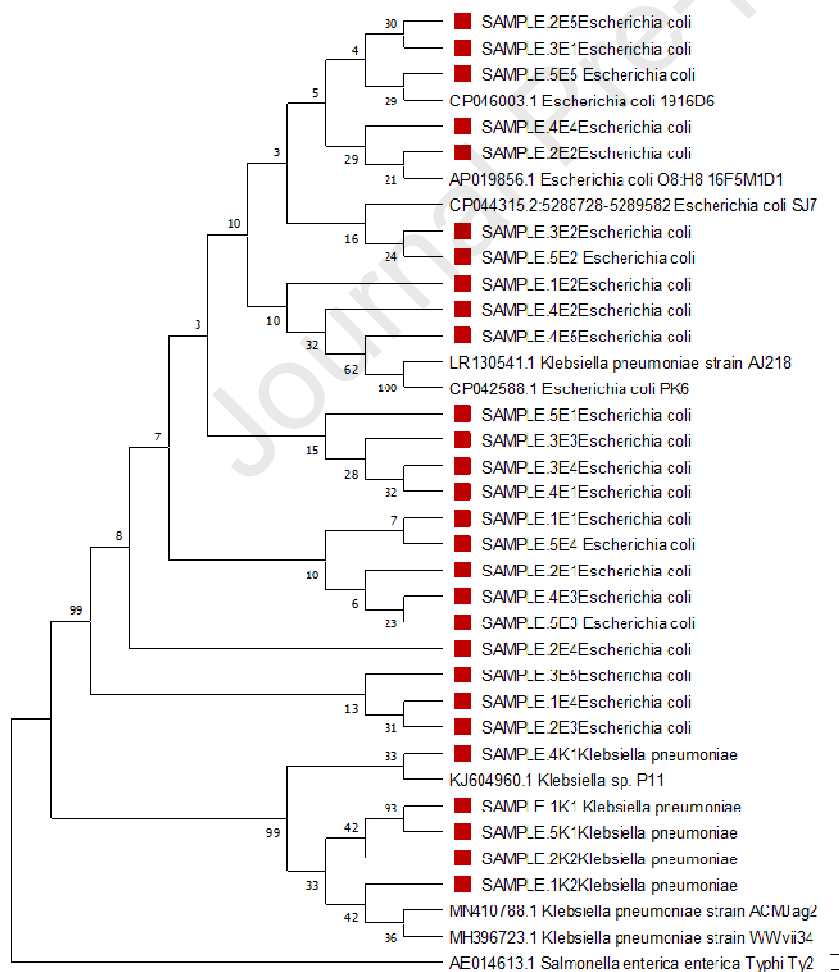


Fig. 2. 16S rDNA phylogeny of selected bacterial isolates (red square dots) showing the similarity of isolated *E. coli* and *Klebsiella* sp. to the selected reference strains.

Antibiotic Susceptibility Test

Out of the 9 major antibiotics tested, *E. coli* and *Klebsiella* sp. exhibited highest resistance to tetracycline. Here, *E. coli* showed 77.27% resistance to tetracycline followed by 13.63% to aztreonam and ciprofloxacin. However, all the *Klebsiella* sp. isolates (100%) were found to be resistant to tetracycline, followed by 60% to aztreonam, 40% to ceftriaxone and gentamycin. All the *E. coli* isolates were sensitive to carbapenem, while *Klebsiella* sp. showed 10% and 20% resistance to ertapenem and imipenem respectively (Fig. 3).

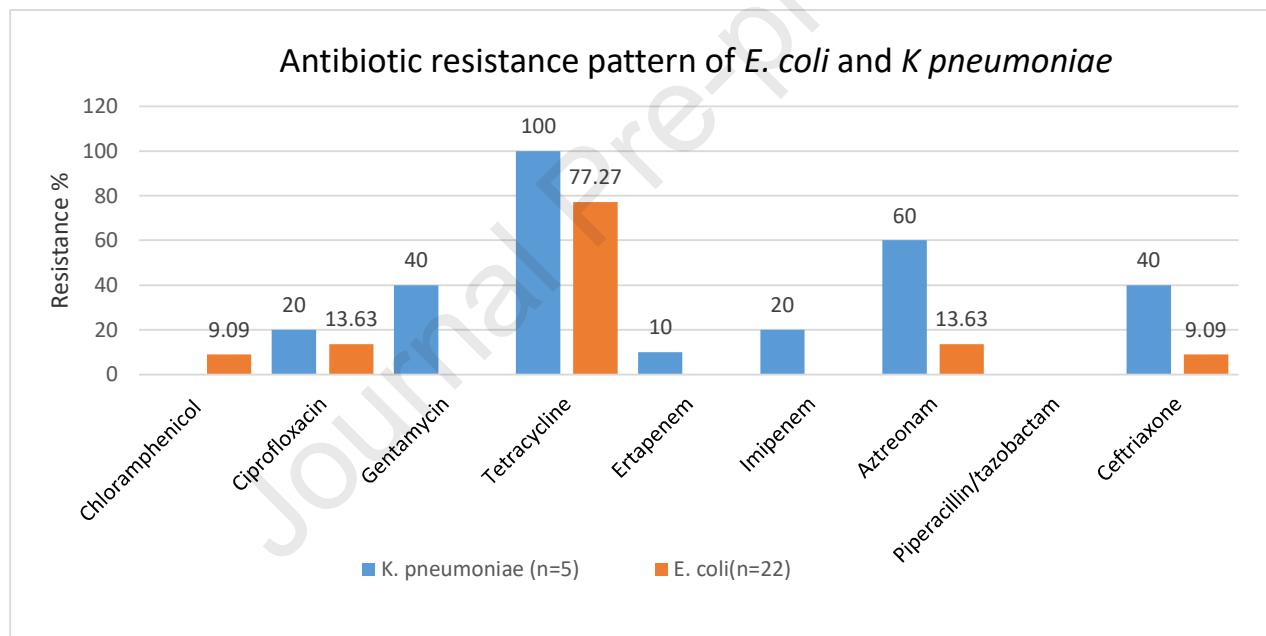


Fig.3. Antimicrobial resistance pattern of *E. coli* and *Klebsiella* sp. isolated in the study.

PCR Amplification of tetracycline resistance gene

The PCR amplification has resulted in the formation of products with size of 501 and 929bp respectively for *tetA* and *tetB* genes. Among the 22 *E. coli* isolates, 17 were positive for *tetA* gene and 3 were positive for *tetB* gene. The *tetA* gene was found to be present in all the *Klebsiella* sp. (n=5) isolates but, no *tetB* gene could be detected. PCR products obtained for *tetA*

and *tetB* genes were sequenced and analysed using The CARD BlastN. All the *tetA* genes from *Klebsiella* and *E coli* shows average of 99.61% identity with the *tetA* sequences of *Shigella sonnei* (ARO:3000165). Further, phylogenetic analysis of both genes with reference sequences confirmed to be *tetA* and *tetB* (Fig. 4 and 5).

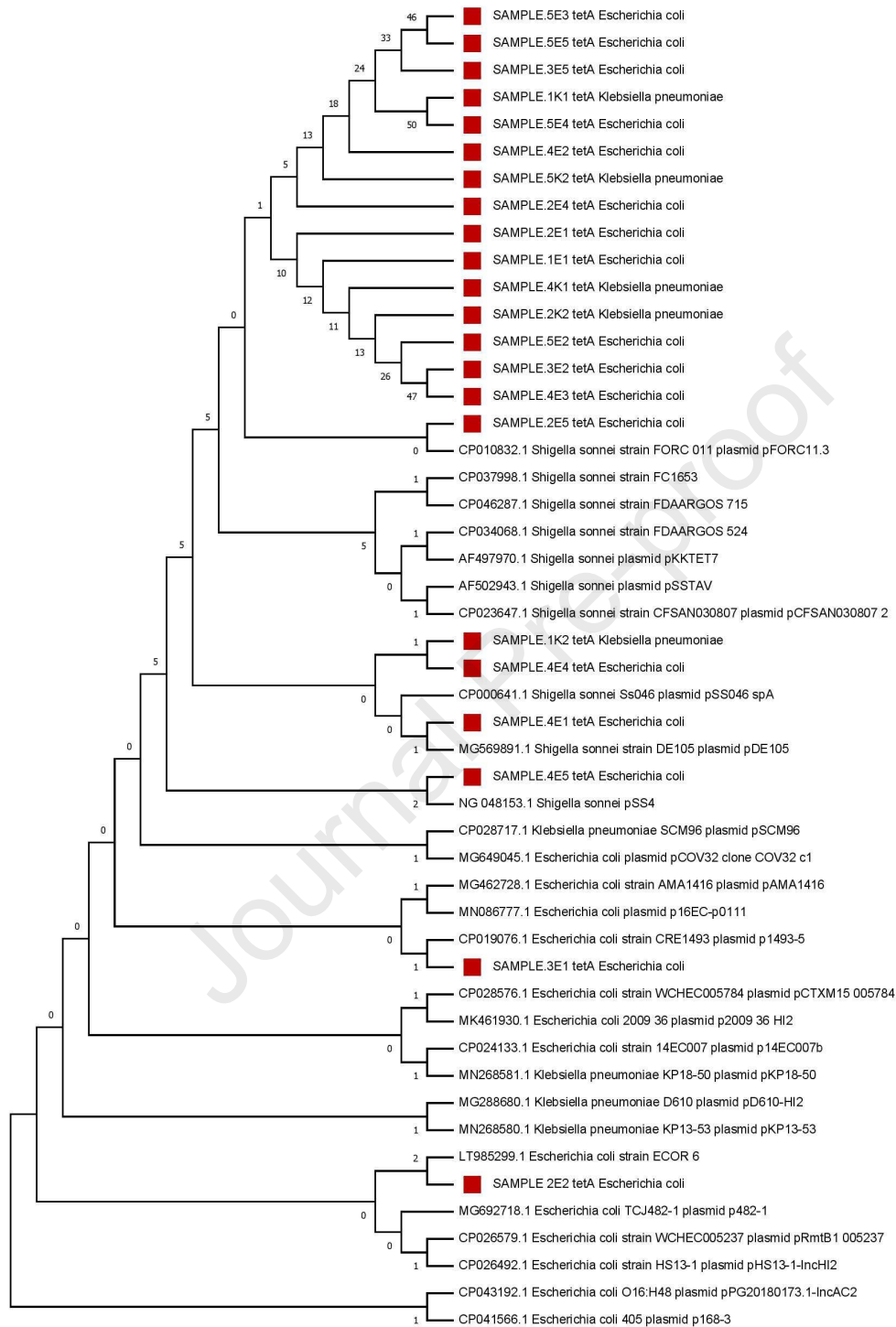


Fig. 4. Phylogenetic analysis showing the evolutionary relationship of identified *tetA* genes from *E. coli* and *K. pneumoniae* with reference sequences.

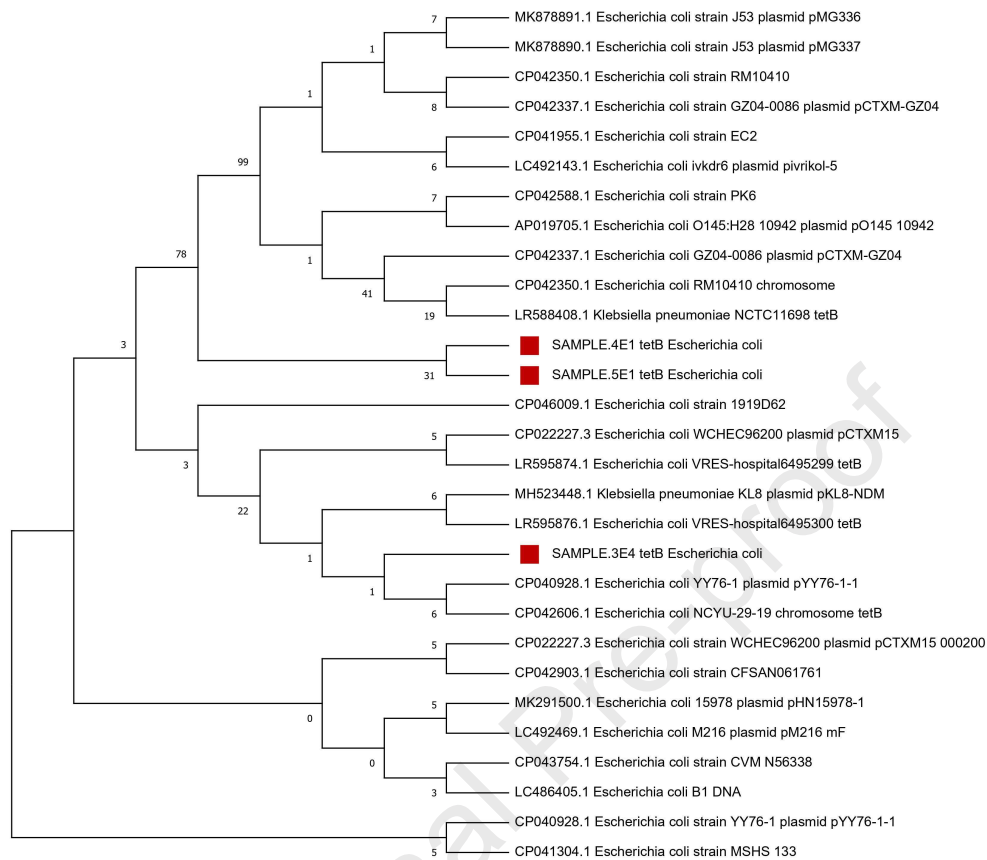


Fig. 5. Phylogenetic analysis showing the evolutionary relationship of identified *tetB* genes from *E. coli* and *K. pneumoniae* with reference sequences.

DISCUSSION

Antimicrobial resistance evolution associated with the livestock sector is an emerging threat to social health and wellbeing. In addition to the use in veterinary medicine, different classes of antibiotics are reported to be added into animal feeds for prophylaxis and growth enhancement[17]. Cattle, pig and poultry are the major sources of meat for human consumption and among these broiler chicken production is the one which requires only minimum time for the growth. The average antibiotic content in chicken feed has been reported to be 148 mg/kg globally[18]. Oxytetracycline, chlortetracycline, lasalocid, neomycin, sulfamethazine and

penicillin are the common antibiotics used in animal feeds. Oxytetracycline and chlortetracycline have been used in almost all categories of animal feeds at concentrations ranging from 50 to 500mg/kg for different purposes. These are used for the prophylactic control of leptospirosis, bacterial enteritis and bacterial pneumonia, protection from *Mycoplasma synoviae*, *Pasteurella multocida*, and *E. coli*[19]. In the current study, 3 out of 5 feed samples were found to have the presence of tetracycline and hence this could be the key driver for resistance development among the isolated *E. coli* and *Klebsiella* sp. The presence of antibiotics in feed and the farm environment may help the *tetA* and *tetB* genes to persist in the microbiome for a long time. The presence of antibiotics even in small amount ensures the sustainability of such resistant genes and may dominantly express in the microbial community.

Even though antibiotic supplemented feed could avoid pathogenic infection for a short time, it can severely affect the healthy microflora of the animal gut and also promotes the selection of resistant strains. The subclinical consumption of antibiotics has also been described to promote the development of resistance mechanisms in microorganisms. Induced resistome formed among the heterogenic gut microbiome will eventually get released through the faeces in to the environment [19]. In this study, 5 different healthy broiler farms from Ernakulum district of Kerala were selected for detailed investigation about resistance buildup in response to feeding with antibiotic supplemented feed. *E.coli* and *Klebsiella* sp. were isolated from the faeces as candidate microorganisms. However, the presence of *Klebsiella* sp. (n=5) were less in the faecal samples when compared with *E. coli* (n=22). *E. coli* and *Klebsiella* sp. are commensal opportunistic pathogens which are common in the intestine of poultry. These pathogens are the causative agents of diseases such as bacterial enteritis, bacterial diarrhea, respiratory and air sac infections in poultry[20].

Developing drug resistance among commensal organisms can favor the development of unresponsiveness towards the antibiotic administrated and hence can increase the mortality and economic loss. From the results of the study, enhanced resistance in all the isolates was observed against the tetracycline which is a common feed additive. Tetracycline is one of the broad spectrum antibiotics active against Gram positive and Gram negative bacteria as well as protozoan parasites, chlamydiae and *Mycobacterium*. Mechanistically, tetracycline inhibits the protein synthesis in bacteria by blocking the binding of aminoacyl-tRNA to the ribosomal

acceptor site[21]. The repeated use of tetracycline by humans in clinical, veterinary and agriculture sectors has resulted in the evolution of microbial resistance. Mainly three microbial resistance mechanisms have been described against the tetracycline and are active efflux pump, ribosomal protection and enzymatic inactivation. The active efflux of tetracycline is the most commonly observed resistance and is plasmid encoded. *tetA* and *tetB* genes codes for tetracycline efflux pumps which comes under the family of major facilitator protein. These efflux systems pumps out the antibiotics and reduce its concentration inside the cell leading to resistance.

In this study, *tetA* gene was found to be common among the isolates, while *tetB* was comparatively less abundant. The CARD BLAST analysis further revealed *tetA* genes in the isolates are 99.61 % percentage identical to the *tetA* genes of *Shigella sonnei* (ARO:3000165). *Shigella sonnei* is a gram negative bacteria comes under the class of Enterobacteriaceae which is also same for *E. coli* and *Klebsiella pneumoniae*. It is previously report that *Shigella sonnei* may have an ability transfer the resistance genes like CTX-M to *Escherichia coli* in the same ecological niche [10]. The presence of *tetA* genes which is genetically related to the *Shigella sp.* may be the result of horizontal gene transfers within the microbiome. However, the *tetB* genes where less abundant and shows similarity to the *E. coli*. Presence of resistance genes in plasmids make it to have high-risk concern as it may be shared through conjugation. *IncP* plasmids have previously been reported to carry AMR with the capacity to conjugate with broad range of Gram negative bacteria such as *Shigella*, *E.coli*, *Klebsiella*, *Pseudomonas* etc.[9,22]. The drug resistant pathogens from livestock farms with plasmid mediated resistance can transfer it easily to the environment [23]. The use of farm manure containing drug resistant pathogens for the agricultural application can be a potential way of disseminating the AMR through crops as well as waterbodies [5].

In the maximum likely-hood phylogeny, 14 *tetA* genes from the 17 formed a distinct phylogenetic clade from the reference sequence (Fig. 4 and 5). This may due to the mutational changes occurred in the genes. Tetracycline resistance mutations have also been reported to develop resistance or reduce the susceptibility to tigecycline which is classified under glycylcyclines family [24,25]. Hence, development of antibiotic resistance in livestock may also cause cross drug resistance, which is critical in medical practices. Here, the healthy poultry feces

collected from all the different farms shows the presence of tetracycline resistance. However, tetracycline antibiotic in feed has been detected only in 3 farms. Further investigations are necessary to identify the administration of tetracycline or tetracycline analogs other than through feed concentrates, which will also contribute to the antibiotic resistance. Recent studies show the transferability of the resistance genes from livestock to humans which is an emerging risk factor for public health[26]. Transfer of tetracycline resistance from the pig farms to the gut flora of animal caretakers have been reported in 2020. Such zoonotic resistance transfer could make the infectious disease treatment difficult[27].

This study emphasizes the chances of development of antimicrobial resistance in healthy poultry farms due to antibiotic supplemented feed. Dissemination of drug resistant bacteria through the feces may contaminate the farm premises as well as the care takers. The results of the study show that addition of antibiotics in animal feed must be prevented in order to control AMR.

CONCLUSION

Antibiotic usage in animal feed for disease prevention and weight gain may cause the evolution of antimicrobial resistance. The healthy gut microbiome may tremendously change and could be replaced by the drug resistant strains. The Gram- negative, *E. coli* and *Klebsiella pneumoniae*, isolated in the study shows high resistance to tetracycline which might have favored due to the addition of this antibiotics as feed additives. Also, the resistance genes are suspected to be accrued through the closely related species *Shigella sonnei* and its continues evolution may cause cross resistance to other antibiotics. The quantification of tetracycline in feed by LC – MS confirmed the presence of tetracycline in collected 3 out of 5 feed samples which supports the observation of resistance buildup. Antibiotics in animal feed should be monitored properly to avoid the resistance development. The prophylaxis and healthy weight gain could be achieved by an antibiotic free diet plan, which includes proper hygiene of the animal sheds, ventilation and nutritious feeding material

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Bibliography

- [1] M. Ferri, E. Ranucci, P. Romagnoli, V. Giaccone, Antimicrobial resistance: A global emerging threat to public health systems, *Crit. Rev. Food Sci. Nutr.* 57 (2017) 2857–2876. <https://doi.org/10.1080/10408398.2015.1077192>.
- [2] Y. Mehdi, M.P. Létourneau-Montminy, M. Lou Gaucher, Y. Chorfi, G. Suresh, T. Rouissi, S.K. Brar, C. Côté, A.A. Ramirez, S. Godbout, Use of antibiotics in broiler production: Global impacts and alternatives, *Anim. Nutr.* 4 (2018) 170–178. <https://doi.org/10.1016/j.aninu.2018.03.002>.
- [3] F. Shariatmadari, Plans of feeding broiler chickens, *Worlds. Poult. Sci. J.* 68 (2012) 21–30. <https://doi.org/10.1017/S0043933912000037>.
- [4] W. Xiong, Y. Wang, Y. Sun, L. Ma, Q. Zeng, X. Jiang, A. Li, Z. Zeng, T. Zhang, Antibiotic-mediated changes in the fecal microbiome of broiler chickens define the incidence of antibiotic resistance genes, *Microbiome*. 6 (2018). <https://doi.org/10.1186/s40168-018-0419-2>.
- [5] H. Dahshan, A.M.M. Abd-Elall, A.M. Megahed, M.A. Abd-El-Kader, E.E. Nabawy, Veterinary antibiotic resistance, residues, and ecological risks in environmental samples obtained from poultry farms, Egypt, *Environ. Monit. Assess.* 187 (2015). <https://doi.org/10.1007/s10661-014-4218-3>.
- [6] S. Schwarz, A. Loeffler, K. Kadlec, Bacterial resistance to antimicrobial agents and its impact on veterinary and human medicine, in: *Adv. Vet. Dermatology*, John Wiley & Sons, Ltd, 2017: pp. 95–110. <https://doi.org/10.1002/9781119278368.ch5.1>.
- [7] A. Bryan, N. Shapir, M.J. Sadowsky, Frequency and Distribution of Tetracycline Resistance Genes in Genetically Diverse, Nonselected, and Nonclinical *Escherichia coli* Strains Isolated from Diverse Human and Animal Sources, *Appl. Environ. Microbiol.* 70 (2004) 2503–2507. <https://doi.org/10.1128/AEM.70.4.2503-2507.2004>.
- [8] M.C. Roberts, Update on acquired tetracycline resistance genes, *FEMS Microbiol. Lett.* 245 (2005) 195–203. <https://doi.org/10.1016/j.femsle.2005.02.034>.

- [9] M. Popowska, A. Krawczyk-Balska, Broad-host-range IncP-1 plasmids and their resistance potential, *Front. Microbiol.* 4 (2013). <https://doi.org/10.3389/fmicb.2013.00044>.
- [10] S.H. Cho, S.Y. Han, Y.H. Kang, Possibility of CTX-M-14 gene transfer from *Shigella sonnei* to a commensal *Escherichia coli* strain of the gastroenteritis microbiome, *Osong Public Heal. Res. Perspect.* 5 (2014) 156–160. <https://doi.org/10.1016/j.phrp.2014.04.007>.
- [11] R. Gavilán, C. Nebot, J. Miranda, Y. Martín-Gómez, B. Vázquez-Belda, C. Franco, A. Cepeda, Analysis of Tetracyclines in Medicated Feed for Food Animal Production by HPLC-MS/MS, *Antibiotics.* 5 (2015) 1. <https://doi.org/10.3390/antibiotics5010001>.
- [12] A.S. Obeng, H. Rickard, O. Ndi, M. Sexton, M. Barton, Antibiotic resistance, phylogenetic grouping and virulence potential of *Escherichia coli* isolated from the faeces of intensively farmed and free range poultry, *Vet. Microbiol.* 154 (2012) 305–315. <https://doi.org/10.1016/j.vetmic.2011.07.010>.
- [13] M.A. Khasru Parvez, M. Marzan, Prevalence of Inhibitor Resistant Beta Lactamase Producing *E. coli* in Human and Poultry Origin of Bangladesh, *J. Bacteriol. Parasitol.* 07 (2016). <https://doi.org/10.4172/2155-9597.1000271>.
- [14] C. Varga, M.T. Guerin, M.L. Brash, D. Slavic, P. Boerlin, L. Susta, Antimicrobial resistance in fecal *Escherichia coli* and *Salmonella enterica* isolates: A two-year prospective study of small poultry flocks in Ontario, Canada, *BMC Vet. Res.* 15 (2019). <https://doi.org/10.1186/s12917-019-2187-z>.
- [15] O.A. Olowe, O.J. Idris, S.S. Taiwo, Prevalence of TET genes mediating tetracycline resistance in *Escherichia coli* clinical isolates in Osun State, Nigeria, *Eur. J. Microbiol. Immunol.* 3 (2013) 135–140. <https://doi.org/10.1556/eujmi.3.2013.2.7>.
- [16] A.G. McArthur, N. Waglechner, F. Nizam, A. Yan, M.A. Azad, A.J. Baylay, K. Bhullar, M.J. Canova, G. De Pascale, L. Ejim, L. Kalan, A.M. King, K. Koteva, M. Morar, M.R. Mulvey, J.S. O'Brien, A.C. Pawlowski, L.J.V. Piddock, P. Spanogiannopoulos, A.D. Sutherland, I. Tang, P.L. Taylor, M. Thaker, W. Wang, M. Yan, T. Yu, G.D. Wright, The comprehensive antibiotic resistance database, *Antimicrob. Agents Chemother.* 57 (2013) 3348–3357. <https://doi.org/10.1128/AAC.00419-13>.

- [17] R. Manishimwe, K. Nishimwe, L. Ojok, Assessment of antibiotic use in farm animals in Rwanda, *Trop. Anim. Health Prod.* 49 (2017) 1101–1106. <https://doi.org/10.1007/s11250-017-1290-z>.
- [18] A. Kumar, A. Patyal, A.. Panda, Sub-therapeutic use of antibiotics in animal feed and their potential impact on environmental and human health: a comprehensive review, *J. Anim. Feed Sci. Technol.* 6 (2018) 15–25. <https://doi.org/http://dx.doi.org/10.21088/jafst.2321.1628.6118.3>.
- [19] F. Granados-Chinchilla, C. Rodríguez, Tetracyclines in Food and Feedingstuffs: From Regulation to Analytical Methods, Bacterial Resistance, and Environmental and Health Implications, *J. Anal. Methods Chem.* 2017 (2017). <https://doi.org/10.1155/2017/1315497>.
- [20] N. Popy, M. Asaduzzaman, M.S. Miah, A. Siddika, M.A. Sufian, M.M. Hossain, Pathological study on the upper respiratory tract infection of chickens and isolation, identification of causal bacteria, *Bangladesh Vet.* 28 (2012) 60–69. <https://doi.org/10.3329/bvet.v28i2.10677>.
- [21] D. Schnappinger, W. Hillen, Tetracyclines: Antibiotic action, uptake, and resistance mechanisms, *Arch. Microbiol.* 165 (1996) 359–369. <https://doi.org/10.1007/s002030050339>.
- [22] J. Madhavi, M. Santoshi, K.R. Meghanadh, A. Jyothy, Prevalence and antibiotic sensitivity of *Staphylococcus aureus* and *Pseudomonas aeruginosa* in middle ear fluids of chronic suppurative otitis media and chronic rhinosinusitis patients undergoing ear surgery, *Int. J. Infect. Dis.* 45 (2016) 101–102. <https://doi.org/10.1016/j.ijid.2016.02.264>.
- [23] M. Usui, T. Shirakawa, A. Fukuda, Y. Tamura, The role of flies in disseminating plasmids with antimicrobial-resistance genes between farms, *Microb. Drug Resist.* 21 (2015) 562–569. <https://doi.org/10.1089/mdr.2015.0033>.
- [24] X. Du, F. He, Q. Shi, F. Zhao, J. Xu, Y. Fu, Y. Yu, The rapid emergence of tigecycline resistance in blaKPC-2 harboring *Klebsiella pneumoniae*, as mediated in vivo by mutation in tetA during tigecycline treatment, *Front. Microbiol.* 9 (2018).

<https://doi.org/10.3389/fmicb.2018.00648>.

- [25] M. Linkevicius, L. Sandegren, D.I. Andersson, Potential of tetracycline resistance proteins to evolve tigecycline resistance, *Antimicrob. Agents Chemother.* 60 (2016) 789–796.
<https://doi.org/10.1128/AAC.02465-15>.
- [26] F. Ma, S. Xu, Z. Tang, Z. Li, L. Zhang, Use of antimicrobials in food animals and impact of transmission of antimicrobial resistance on humans, *Biosaf. Heal.* (2020).
<https://doi.org/10.1016/j.bsheal.2020.09.004>.
- [27] Q. Luo, Y. Wang, Y. Xiao, Prevalence and transmission of mobilized colistin resistance (mcr) gene in bacteria common to animals and humans, *Biosaf. Heal.* (2020).
<https://doi.org/10.1016/j.bsheal.2020.05.001>.

Highlights

- Tetracycline antibiotics were detected in poultry feeds.
- Tetracycline resistant bacteria were isolated from the feces of the commercial broilers.
- *tetA* and *tetB* genes were detected from the resistant bacterial isolates.

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To

The Editor
Microbial Pathogenesis

Sub: Conflict of interest

This is to state that the authors have no conflict of interest in the article titled “**Healthy broilers disseminate antibiotic resistance in response to tetracycline input in feed concentrates**” .

Thanking You

Sincerely,
Dr. Radhakrishnan EK